

Sunny S. Lou, M.D. Ph.D.

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EDUCATION

2021- 2022	Barnes Jewish Hospital / Washington University Fellowship, Adult Cardiothoracic Anesthesiology	St Louis, MO
2017- 2021	Barnes Jewish Hospital / Washington University Residency, Anesthesiology Internship, Anesthesiology	St Louis, MO
2009- 2017	Stanford University School of Medicine M.D., Ph.D., Chemical and Systems Biology	Stanford, CA
2005- 2009	Massachusetts Institute of Technology S.B., Biology	Cambridge, MA

LICENSES AND CERTIFICATION

2024- present	Board Certification, Adv. Perioperative Transesophageal Echocardiography
2023- present	Board Certification, American Board of Anesthesiology
2017- present	Missouri State Medical License

HONORS AND AWARDS

2024	1st place – Best Clinical Abstract , World Congress for Anaesthesia
2023	Editorial Fellow , British Journal of Anesthesia
2022	Mentored Research Award , International Anesthesia Research Society
2022	Outcomes Research Fellow , Multicenter Perioperative Outcomes Group
2022	Alpha Omega Alpha Honor Society
2021	Team Award for Quality Improvement, Periop Blood Mgmt, BJC Healthcare
2019	Top 10% , American Board of Anesthesiology BASIC Exam
2009	Phi Beta Kappa Honor Society
2008	Barry M. Goldwater Scholarship

POSITIONS

2024- present	Assistant Professor , Washington University Dept of Anesthesiology Divisions of Cardiothoracic Anesthesiology and Clinical Research Affiliate, Institute for Informatics
2022-2024	Instructor , Washington University Dept of Anesthesiology
2017-2022	Resident/Fellow , Washington University Dept of Anesthesiology NIH T32 appointee with mentor: Thomas Kannampallil, PhD Clinical informatics, clinical decision support, machine learning for healthcare
2017	Research Fellow , Google Health Mentor: Ming Jack Po, MD PhD Product design for artificial intelligence in medicine
2010-2017	Doctoral Research , Stanford University Mentor: Julie Theriot, PhD Quantitative microscopy, biophysics, genomics to study cell migration.

RESEARCH SUPPORT

K23 HL166880 07/15/2024-06/30/2028
NIH/NHLBI, Role: PI Annual Directs: \$160,749
Intelligent Clinical Decision Support for Perioperative Blood Management

R01 HS029020 09/30/2022-07/31/2027
Agency for Healthcare Research and Quality, Role: Co-Investigator Annual Directs: \$281,814
Integrating real-time clinical activity and behavioral responses for characterizing cognitive load and errors (IGNITE)

EHR-Use Metrics Research 06/01/2022-12/31/2023
American Medical Association Foundation, Role: PI Annual Directs: \$50,000
Metrics for EHR Secure Chat Use

SERVICE

- 2021- present **Barnes Jewish Hospital Perioperative Blood Management Committee**
Developed real-time reports to measure perioperative blood delivery and usage.
Led quality improvement efforts to reduce blood product waste and to improve the efficiency of blood delivery to the operating rooms.
- 2023- present **Department of Anesthesia Data Governance Committee**
Oversaw requests for electronic health record data for research, quality improvement, clinical operations, and billing.
- 2024- present **Scientific Review Board Member.** Multicenter Perioperative Outcomes Group
Proposal Review Committee, Anesthesia Patient Safety Foundation Committee on Scientific Evaluation

SCIENTIFIC CONTRIBUTIONS

Clinical Decision Support for Perioperative Transfusion. Developed a machine-learning model to predict the risk of red cell transfusion during surgery, and evaluated its performance compared to standard of care approaches using clinically meaningful benchmarks. Externally validated model performance at many hospital systems across the US. Implemented the model as a clinical decision support tool embedded within the EHR, and led a human-centered design process to integrate the tool within clinical workflow.

- a. **Lou SS**, Liu H, Lu C, Wildes TS, Hall BL, Kannampallil T. Personalized surgical transfusion risk prediction using machine learning to guide preoperative type and screen orders. *Anesthesiology*. 2022 Jul 1;137(1):55-66. PMID: 35147666. doi: 10.1097/aln.0000000000004139
- b. **Lou SS**, Dewey MM, Bollini ML, Harford DR, Ingold C, Wildes TS, Stevens TW, Martin JL, Grossman BJ, Kangrga I (2023) Reducing perioperative red blood cell unit issue orders, returns, and waste using failure modes and effects analysis. *Transfusion*. 2023 Apr;63(4):755-762. doi: 10.1111/trf.17275

- c. Yang P, Zijlstra EP, Hall BL, Gregory SH, Jackups R, Li J, Abraham J, **Lou SS** (2024) Challenges in reliable preoperative blood ordering: a qualitative interview study. *Transfusion*. doi: 10.1111/trf.18012
- d. **Lou SS**, Kumar S, Goss CW, Avidan MS, Kheterpal S, Kannampallil T. Multicenter Validation of a Machine Learning Model for Surgical Transfusion Risk at 45 US Hospitals. *JAMA Network Open*. 2025 Jun 2;8(6):e2517760. doi: 10.1001/jamanetworkopen.2025.17760

Measurement of Clinician Workload and Cognitive Burden. All activities performed on an EHR are recorded in audit log files to monitor access to protected patient health information. I developed novel methods that use audit log data for large-scale measurement of clinical workload and work activities. Using these methods, I have evaluated the relationship between clinical workload, EHR use, cognitive burden, medication error, and reimbursement. These findings have implications for the development and evaluation of interventions designed to reduce clinician burden and burnout.

- a. **Lou SS**, Lew D, Harford DR, Lu C, Evanoff BA, Duncan JG, Kannampallil T. Temporal associations between EHR-derived workload, burnout, and errors: a prospective cohort study. *Journal of General Internal Medicine*. 2022 Jul;37(9):2165-2172. doi: 10.1007/s11606-022-07620-3
- b. **Lou SS**, Kim S, Harford D, Warner BC, Payne PRO, Abraham J, Kannampallil T. Effect of Clinician Attention Switching on Workload and Wrong-Patient Errors. *British Journal of Anaesthesia*. 2022 Jul;129(1):e22-e24. doi: 10.1016/j.bja.2022.04.012.
- c. **Lou SS**, Baratta LR, Lew D, Harford D, Avidan MS, Kannampallil T. Comparison of anesthesia clinical workload estimated from electronic health record documentation vs billed Relative Value Units, *JAMA Network Open*. 2023 Aug 1;6(8):e2328514. doi:10.1001/jamanetworkopen.2023.28514
- d. Kim S, Warner BC, Lew D, **Lou SS**, Kannampallil T. Measuring cognitive effort using tabular transformer-based language models of electronic health record-based audit log sequences. *J Am Med Inform Assoc*. 2024 Oct 1;31(10):2228-2235. doi: 10.1093/jamia/ocae171.

Impact of EHR-integrated Secure Messaging. Communication between clinicians is essential for effective and safe patient care. Secure text messaging applications, which enable asynchronous closed loop messaging between clinicians, have emerged as a popular form of communication over the past decade. Despite widespread adoption, the implications of secure messaging use on clinician workflow and communication practices remain poorly understood. I have characterized the patterns of secure messaging use and its associations with clinician workload, cognitive burden, and patient safety. These results have implications for the design of novel communication technologies.

- a. Baratta LR, Harford D, Sinsky CA, Kannampallil T, **Lou SS** (2023) Characterizing the patterns of EHR-integrated secure messaging use: A cross-sectional study. *J. Med. Internet Res*. doi: 10.2196/48583.
- b. **Lou SS**, Lew D, Baratta LR, Eiden E, Sinsky CA, Kannampallil T (2024) Secure Messaging and Telephone Use for Clinician-to-Clinician Communication. *JAMA Network Open*. doi: 10.1001/jamanetworkopen.2024.1778
- c. **Lou SS**, Lew D, Xia L, Baratta L, Eiden E, Kannampallil T (2024) Secure messaging use and wrong-patient ordering errors among inpatient clinicians. *JAMA Network Open*. doi: 10.1001/jamanetworkopen.2024.47797
- d. Lew D, Baratta LR, Xia L, Eiden E, Sinsky CA, Kannampallil T, **Lou SS**. (2025) Association of EHR-Integrated Secure Messaging Use with Clinician Workload and Attention Switching. *Journal of General Internal Medicine*. 2025 Mar 14. doi: 10.1007/s11606-025-09466-x.